A SUFFICIENTLY DIVISIBLE VERONESE SUBRING OF A FINITELY GENERATED GRADED ALGEBRA IS GENERATED IN DEGREE ONE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Homogeneous generators of a graded algebra over its degree-zero part. Used at the start of the counted proof (lines 13507-13509). Replaces the generators by homogeneous ones of positive degree, which is what makes the monomial description available.
- The pigeonhole argument on exponent vectors. Invoked in the middle of the counted proof (lines 13513-13516). Shows that a monomial of degree N m with N at least r has one exponent at least m/d_i.

Lemma 0.1. *Let S be a graded ring, which is finitely generated over S_0. Then for all sufficiently divisible d the algebra $S^{(d)}$ is generated in degree 1 over S_0.*

Proof. Say S is generated by $f_1, \ldots, f_r \in S$ over S_0. After replacing f_i by their homogeneous parts, we may assume f_i is homogeneous of degree $d_i > 0$. Then any element of S_n is a linear combination with coefficients in S_0 of monomials $f_1^{e_1} \ldots f_r^{e_r}$ with $\sum e_i d_i = n$. Let m be a multiple of $\text{lcm}(d_i)$. For any $N \geq r$ if

$$\sum e_i d_i = Nm$$

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then for some i we have $e_i \geq m/d_i$ by an elementary argument. Hence every monomial of degree Nm is a product of a monomial of degree m, namely f_i^{m/d_i}, and a monomial of degree $(N-1)m$. It follows that any monomial of degree nrm with $n \geq 2$ is a product of monomials of degree rm. Thus $S^{(rm)}$ is generated in degree 1 over S_0. $\square$